



A LOW-COST AIR-FREE TRANSPORT CONTAINER FOR OXYGEN-SENSITIVE SAMPLES

MECHANICAL & MATERIALS ENGINEERING

TOOLS & METHODS:

ANSYS Fluent (transient two-phase CFD, realizable k- ϵ turbulence model) · SOLIDWORKS Flow Simulation · SOLIDWORKS CAD & FEA · additive manufacturing of fixtures and seals · glovebox sample handling below 1 ppm O₂ · colorimetric oxygen indicators · accelerated degradation testing against FLiNaK molten salt and PbLi alloy.

EXECUTIVE SUMMARY:

Lithium metal, uranium, FIB-prepared TEM lamellae and molten-salt reactor fuels such as FLiNaK oxidize within seconds to minutes in air, invalidating experiments and destroying samples that can cost tens of thousands of dollars to prepare. Gloveboxes protect samples but are not portable, and commercial inert-transfer holders are expensive and typically hold inert conditions for less than 48 hours.

This programme advanced a compact, low-cost container toward production. Computational fluid dynamics established how fast and how long to purge the vessel with argon, a clamp redesign and purge-confirmation attachment made it operable by a gloved hand, and two-month storage trials showed FLiNaK preserved with no visible degradation.

KEY RESULTS:

- **0.05 m/s** optimal argon inlet velocity from a four-case CFD sweep (0.5 → 0.01 m/s): full purge in ~4 min with no trapped air.
- **< 20 ppm O₂ after 240 s** of purging, predicted by 3D flow simulation of the whole container.
- **Two months of FLiNaK storage** with no visible degradation; an air-exposed control decayed within a week.
- Holds inert conditions **> 4 months at ~1/10 the cost** of commercial transfer holders; fits standard TEM holder ports.
- Screwdriver-tightened ring clamp replaced by a **linear-force bar clamp** validated with FEA.
- Root cause of the one observed seal failure isolated to **septum lubrication**, and closed out procedurally.

INTRODUCTION:

Some materials form an oxide layer at oxygen concentrations of 1 ppm. Uranium oxidizes within seconds of air exposure, plutonium within minutes, and FLiNaK (LiF–NaF–KF) reacts with O₂ and moisture to form HF and hydroxides that destroy its crystal structure. Battery, nuclear, 2D-materials and semiconductor researchers therefore need to move samples from a glovebox to TEM, SEM, XPS or SIMS instruments without ever touching air.

The available options force a compromise. Gloveboxes and Schlenk lines are stationary, shared and prone to cross-contamination; commercial transfer holders are portable but cost far more than most laboratories can dedicate to routine storage, and leak measurably after roughly 48 hours. An injection-molded container closes that gap, provided it can be purged reliably and sealed by a gloved hand inside the box.

TEST SETUP:

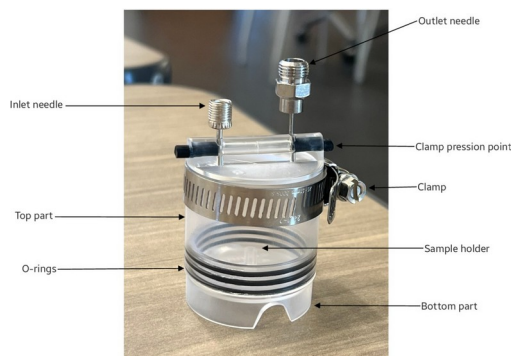


FIG 1: Container assembly. The inlet needle sits lower than the outlet so heavier argon fills from the bottom while air is displaced upward; three O-rings seal the two halves.

Purge-flow modeling. Of the four inert-gas purging strategies, namely dilution, displacement, pressure cycling and evacuation, only dilution/displacement suits a vessel that is neither vacuum- nor pressure-rated and whose inlet and outlet share one face. A cross-section with needles inserted was meshed in ANSYS with structured face meshing refined at the needle openings; a four-mesh independence study converged at 0.25 mm. The transient solver used gravity, the realizable k- ϵ turbulence model with scalable wall functions, and a two-phase argon/air formulation initialized to 100% air. Four inlet velocities (0.5, 0.1, 0.05 and 0.01 m/s) were compared for settling time and residual air. Because the planar model left artificial trapped-air regions, the study was repeated in 3D in SOLIDWORKS Flow Simulation.

Mechanical redesign. With the molded body fixed, improvements targeted the surrounding hardware. A ring

clamp that required a screwdriver inside the glovebox was replaced by a bar clamp applying force only at the two points that compress the septum, with a curved seat and needle holes; an outlet-needle attachment confirms visually that gas is flowing during a purge. Two further concepts were prototyped: a twist-open acme-thread grip that drives the bottom part free of its O-rings, and a clamp-free seal using two 3D-printed pistons carrying three O-rings each.

Storage trials. Samples were loaded in a glovebox below 1 ppm O₂, purged at ≤ 2 psi for ten minutes, clamped and removed to ambient air alongside exposed controls. FLiNaK was the primary indicator because its degradation is visible; a PbLi alloy with an internal colorimetric oxygen indicator provided a cross-check.

RESULTS:

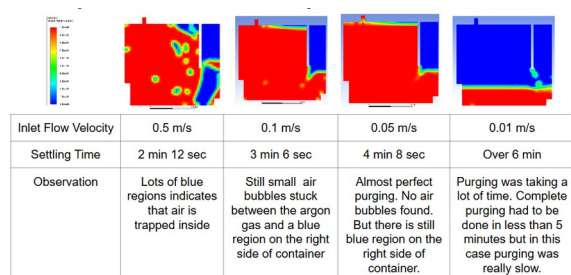


FIG 2: ANSYS Fluent results for the four inlet velocities (blue indicates residual air). At 0.05 m/s the cavity purges almost completely in about four minutes; faster inlets trap bubbles against the wall, and slower ones exceed six minutes.

The sweep identified 0.05 m/s as the operating point: higher flow rates entrained air pockets that persisted past the nominal settling time, while 0.01 m/s doubled purge duration without improving the endpoint. The 3D model placed oxygen below 20 ppm within four minutes.



FIG 3: FLiNaK after two months sealed in the container (left) versus the air-exposed control (right), a moist paste within three days.

The container held FLiNaK visibly unchanged for two months while the control decayed within days. The PbLi trial failed, and that failure produced the most actionable finding of the programme. The oxygen indicator shifted from pink to purple as soon as the container left the glovebox, signalling 1,000 to

10,000 ppm O₂, and an oxide film formed on the alloy. Inspection traced the leak to dried lubricant on the septum cords: the septum compressed laterally rather than vertically when clamped and never sealed the needle ports. The correction is procedural, and re-lubrication before every use is now a required step. A second constraint emerged from the piston concept, where sharp corners in the molded needle cavity cut the O-rings as the pistons travelled; that approach requires a mold change rather than an add-on part.

APPLICATIONS:

A container that costs a fraction of commercial holders, fits a standard TEM specimen-holder port and holds an inert atmosphere for months makes air-free workflows accessible to any laboratory with a glovebox: lithium-metal battery development, nuclear fuel and U–Mo alloy characterization, FIB-prepared lamellae, and air-sensitive biological and nanomaterial samples. The modular architecture accepts customized needles, sample holders and optional oxygen-monitoring modules, extending it to industrial quality control and materials-supply chains. The path to production is defined: optimize seal manufacturing, integrate an oxygen sensor, add a leak-rate test, and cut mold tooling for the revised clamp.

DISCUSS THIS WORK:

This paper summarises one project. If the same problem appears in your programme, I am happy to walk through the method, the data behind it, and how it would map onto your hardware. emincherif@gmail.com